Assignment 03: Insight Finding & Statistical Inference

From Visual Patterns to Testable Hypotheses

Module: Introduction to Machine Learning

Instructions

- **Type:** Interpretative Analysis and Statistical Reasoning.
- **Goal:** This assignment tests your ability to translate visual patterns (charts) into meaningful statistical insights. You must use concepts like Central Tendency (Mean, Median, Mode) and Dispersion (Variance) to back up your visual claims.
- **Context:** You are a Data Analyst at a Retail Company. You have generated several plots, and now you must explain what they mean to the stakeholders.

Case Study: Retail Sales Analysis

Your dataset $\mathcal{D}$ contains sales transactions from 50 stores.

- x_1: **Daily Sales** (USD)
- x_2: **Customer Footfall** (Count)
- x_3: **Store Type** (A, B, C)
- x_4: **Promotion Active** (Yes/No)

Part 1: Distribution Analysis (Histogram & Box Plot)

Q1. The Skewed Histogram (Daily Sales): You plot a Histogram of x_1 (Daily Sales) and observe a long tail to the right (Positive Skew).

- **Visual Insight:** What does this "tail" represent in the context of retail business? Are these "outliers" or "VIP customers"?
- **Statistical Inference:** In a positively skewed distribution, compare the likely positions of the Mean, Median, and Mode. Which metric would you report to the CEO as "typical daily sales" to avoid over-optimism?

Q2. The Box Plot Comparison (Sales by Store Type): You create a Box Plot of Sales (x_1) grouped by Store Type (x_3).

- **Interpreting Spread:** Store Type A has a much longer "box" (Interquartile Range) than Store Type B. What does this imply about the **Variance** or consistency of performance in Type A vs. Type B?
- **Hypothesis Generation:** Formulate a statistical hypothesis about the stability of revenue. (e.g., "Is the variance of Type A significantly different from Type B?")

Part 2: Relationship & Trends (Scatter & Bar Charts)

Q3. Scatter Plot (Footfall vs. Sales): You plot Footfall (x_2) on the x-axis and Sales (x_1) on the y-axis.

- **Pattern Recognition:** You see a positive linear trend but the points ”fan out” (get wider) as Footfall increases (Heteroscedasticity). What does this mean for prediction accuracy at high footfall levels?
- **Outlier Detection:** You spot a point with High Footfall but Low Sales. Propose a business reason for this anomaly (e.g., ”Window Shoppers” or ”System Failure”).

Q4. Countplot vs. Bar Chart (Promotion Analysis):

- **Countplot:** Shows there are 1000 days with ”No Promotion” and only 100 days with ”Promotion”. What is this phenomenon called (Class Imbalance)?
- **Bar Chart:** Shows the *Average* Sales is \$5000 with Promotion vs \$2000 without.
- **Critical Question:** Is the higher average solely due to the promotion, or could the small sample size (100 days) be introducing noise? How would you verify this using **Standard Deviation** or Error Bars?

Part 3: Correlation & Multivariable Analysis (Heatmap)

Q5. Heatmap Correlation Matrix: You generate a heatmap showing Pearson correlation coefficients (r) between features.

- **Interpretation:** You find $r(x_1, x_2) = 0.85$ (Sales vs Footfall) and $r(x_1, x_4) = 0.60$ (Sales vs Promo). Which factor is a stronger driver of revenue?
- **Collinearity Risk:** You notice highly correlated features (e.g., Footfall and Number of Transactions, $r = 0.95$). Why might including both in a Linear Regression model cause problems?

Part 4: Conclusion (HOTS)

Q6. Executive Summary: Synthesize your findings.

- Based on the high variance in Store A (Box Plot) and the skewed sales distribution (Histogram), recommend a strategy. Should the company focus on raising the *average* sales or reducing the *inconsistency* (variance)?
- Justify your recommendation using the statistical concepts of Risk (Variance) and Reward (Mean).